

Performance Testing

Date	15 March 2026
Team ID	SWTID-2026-7446
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models

Maximum Marks 5 Marks

Performance testing evaluates how the ComicCraft system behaves under expected and peak load conditions. The application uses FastAPI, Gemini Flash/Pro, Stable Diffusion, Jinja2, and PDF export.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter / Locust / k6 / Postman (to be selected)
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	/generate, /generate-comic/json, /test-image
Test Environment	Local FastAPI server (127.0.0.1:8000)
Test Date	_____

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome	Actual Result
1	Generate a complete comic using /generate with story prompt, character, setting, tone and art style.	To be set	To be set	Comic preview generated successfully without errors.	_____
2	Generate a comic through /generate-comic/json and receive layout data and PDF path.	To be set	To be set	Valid JSON response and PDF export path returned.	_____
3	Test image generation independently through /test-image using a custom prompt.	To be set	To be set	Comic-style image generated and saved successfully.	_____
4	Spike test on comic generation to observe behavior during sudden concurrent requests.	To be set	To be set	System handles increased load gracefully with acceptable errors.	_____

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	To be measured	_____	Measure using selected performance tool.
2	Response Time (Max)	< 5 seconds	To be measured	_____	Measure peak response time.
3	Throughput (Req/sec)	To be defined during test	To be measured	_____	Record successful requests per second.
4	Error Rate	< 1%	To be measured	_____	Record failed requests / total requests.
5	CPU Utilization	< 80%	To be measured	_____	Monitor during load test.
6	Memory Utilization	< 80%	To be measured	_____	Monitor during load test.

Step 4: Observations & Analysis

- Comic generation involves multiple AI stages: Gemini Flash outline generation, Gemini Pro story generation, Stable Diffusion image generation, layout building and PDF export.
- The /generate route executes the complete workflow and is therefore the main end-to-end performance target.
- The /generate-comic/json endpoint provides an API-based path for performance testing.
- The /test-image endpoint can be tested separately to isolate image-generation performance.
- Actual response time, throughput, CPU, memory and error-rate values must be captured from the selected performance-testing tool.

Step 5: Screenshots / Evidence

Attach screenshots of the performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

End of Performance Testing Document